

# J. Zachary Gazak

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## Research Interests

Vision-language model interpretability, spatial reasoning in multimodal models, computer vision, autonomous sensing systems, object detection and density estimation, Bayesian uncertainty quantification, space domain awareness, astronomical instrumentation, stellar spectroscopy.

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## Education

**Ph.D., Astronomy** · University of Hawai'i at Mānoa · 2014 Advisor: Rolf-Peter Kudritzki. Thesis on quantitative near-infrared spectroscopy of red supergiant stars as extragalactic metallicity probes.

**M.S., Astronomy** · University of Hawai'i at Mānoa · 2010

**B.S., Physics** · Massachusetts Institute of Technology · 2006

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## Professional Experience

**Principal Investigator & Astronomer** · Space Systems Command (SSC), Kihei, HI · 2020–Present PI of the Machine Intelligence for Space Superiority lab at U.S. Space Systems Command. Lead a 10-person, multi-organization research team (KBR, Boeing, EOS, MIT, Virginia Tech, Macquarie University, UNSW) developing autonomous ML systems for operational deployment in high-stakes space domain awareness environments. Designed and deployed novel sensing modalities (spectroscopic, neuromorphic, multispectral) with end-to-end pipelines from data collection through model training to operational deployment across distributed autonomous sensor networks.

**Independent Researcher** · Sidereal LLC, Haiku, HI · 2024–Present Independent ML research targeting top-tier venues (NeurIPS, ICLR). Current program: diagnosing and mechanistically characterizing spatial reasoning failures in vision-language models via the Sophon benchmark and Frozen Capability Audit framework.

**Research Physicist** · Air Force Research Laboratory (AFRL), Kihei, HI · 2019–2020 Founded a new ML research program for space domain awareness. Built end-to-end simulation, training, and validation pipelines for learned classification from sensor data. Revitalized legacy sensor hardware into high-yield operational platforms through software modernization.

**Technology Co-Founder** · Alfie Hospitality, Sydney, Australia · 2018–2019 Designed, built, and deployed full-stack backend services, mobile applications (iOS/Android), and venue-facing web tools for commercial hospitality operations.

**Venture Capital Associate** · H2 Ventures, Sydney, Australia · 2016–2018 Directed accelerator programs; mentored early-stage fintechs including Weel (DiviPay, \$20M Series A), Forage (\$25M Series B), and Spriggy (\$35M Series B, NAB Ventures).

**Data Scientist & Program Director** · Insight Data Science, Palo Alto, CA · 2014–2016 Led fellowship programs training PhDs for industry data science roles. Founded Insight's internal data science team supporting rapid organizational growth.

**Graduate Research Assistant & Postdoctoral Researcher** · University of Hawai‘i at Mānoa · 2008–2014 Developed automated spectroscopic analysis techniques for extragalactic red supergiant stars. Published first-author papers in leading astrophysics journals. Awarded competitive observing time on Keck, Subaru, and IRTF telescopes.

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## **Publications**

2,200+ citations across 55+ publications (15 first-authored). Full list: [Google Scholar](#).

### **Accepted / Submitted**

1. **J. Z. Gazak**, R. Swindle, J. Fletcher, “Sophon: A Procedural Diagnostic for Spatial Reasoning in Vision-Language Models,” NeurIPS 2026, Datasets & Benchmarks Track (accepted).
2. **J. Z. Gazak**, R. Swindle, L. Spitler, J. Fletcher, “AllClear: Blind All-Sky Astrometric Calibration and Directional Sky-Transparency Maps,” submitted to Publications of the Astronomical Society of the Pacific (2026).

### **Peer-Reviewed Journal & Conference Papers (First-Authored)**

3. **J. Z. Gazak**, R. Swindle, S. Morales, K. Iott, E. Blackhurst, J. Fletcher, “Rapid Deployment, Calibration, and Training of Optical Observatories for Space Domain Awareness,” Journal of the Astronautical Sciences (JASS), 2026 (accepted).
4. **J. Z. Gazak**, L. Fisher, M. Phelps, R. Swindle, L. Baruela, J. Fletcher, “SENPAl: Sidereal Enriched Rate-Track Astrometry in Deep Imagery of Solar System Bodies,” Publications of the Astronomical Society of the Pacific, 2026. DOI: [10.1088/1538-3873/ae2b35](https://doi.org/10.1088/1538-3873/ae2b35).
5. **J. Z. Gazak**, I. McQuaid, R. Swindle, M. Phelps, J. Fletcher, “SpectraNet: Learned Recognition of Artificial Satellites from High Contrast Spectroscopic Imagery,” IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2022, pp. 2403–2411.
6. **J. Z. Gazak**, R.-P. Kudritzki, C. Evans, L. Patrick, B. Davies, M. Bergemann, B. Plez, F. Bresolin, R. Bender, M. Wegner, A. Z. Bonanos, S. J. Williams, “Red Supergiants as Cosmic Abundance Probes: The Sculptor Galaxy NGC 300,” The Astrophysical Journal, 805, 182, 2015.
7. **J. Z. Gazak**, B. Davies, N. Bastian, R.-P. Kudritzki, M. Bergemann, B. Plez, C. Evans, L. Patrick, F. Bresolin, E. Schinnerer, “A New Method for Measuring Metallicities of Young Super Star Clusters,” The Astrophysical Journal, 787, 142, 2014.
8. **J. Z. Gazak**, B. Davies, R.-P. Kudritzki, M. Bergemann, B. Plez, “Quantitative Spectroscopic J-band Study of Red Supergiants in Perseus OB-1,” The Astrophysical Journal, 788, 58, 2014.
9. **J. Z. Gazak**, N. Bastian, R.-P. Kudritzki, A. Adamo, B. Davies, B. Plez, M. A. Urbaneja, “The Red Supergiants in the Super Star Clusters of M83,” Monthly Notices of the Royal Astronomical Society Letters, 430, L35, 2013.
10. **J. Z. Gazak**, J. A. Johnson, J. Tonry, D. Dragomir, J. Eastman, A. W. Mann, E. Agol, “Transit Analysis Package: An IDL Graphical User Interface for Exoplanet Transit Photometry,” Advances in Astronomy, 2012. (100+ citations)

### **Proceedings & Technical Reports (First-Authored)**

11. **J. Z. Gazak**, R. Swindle, S. Morales, M. Phelps, K. Iott, E. Blackhurst, J. Fletcher, “Rapid Deployment, Calibration, and Training of Optical Observatories for Space Domain Awareness,” AMOS Conference, 2025. DOI: [10.64861/XGPY2184](https://doi.org/10.64861/XGPY2184).

12. **J. Z. Gazak**, R. Swindle, M. Phelps, J. Fletcher, "Simultaneous Detection, Recognition, and Localization of Geosynchronous Satellites from Ground Based Imagery," AMOS Conference, 2023.
13. **J. Z. Gazak**, R. Swindle, M. Phelps, I. McQuaid, J. Fletcher, "Learned Satellite Identification Using Low-Resolution Spectropolarimetry," Proc. SPIE 12112, 2022.
14. **J. Z. Gazak**, I. McQuaid, B. Wolfson, J. Fletcher, "Incremental Learning of Novel Resident Space Object Spectral Fingerprints," AMOS Conference, 2021.
15. **J. Z. Gazak**, J. Fletcher, R. Swindle, I. McQuaid, "Exploiting Spatial Information in Raw Spectroscopic Imagery using Convolutional Neural Networks," AMOS Conference, 2020.

### **Selected Co-Authored Publications**

- M. Bergemann, R.-P. Kudritzki, **J. Z. Gazak**, B. Davies, B. Plez, "Red Supergiant Stars as Cosmic Abundance Probes: NLTE Effects in J-band Magnesium Lines," *The Astrophysical Journal*, 804, 113, 2015.
- C. Lardo, B. Davies, R.-P. Kudritzki, **J. Z. Gazak**, C. J. Evans, L. R. Patrick, M. Bergemann, B. Plez, "Red Supergiants as Cosmic Abundance Probes: The First Direct Metallicity Determination of NGC 4038 in the Antennae," *The Astrophysical Journal*, 812, 160, 2015.
- L. R. Patrick, C. J. Evans, B. Davies, R.-P. Kudritzki, **J. Z. Gazak**, M. Bergemann, B. Plez, A. M. N. Ferguson, "Red Supergiant Stars as Cosmic Abundance Probes: KMOS Observations in NGC 6822," *The Astrophysical Journal*, 803, 14, 2015.
- B. Davies, R.-P. Kudritzki, **J. Z. Gazak**, B. Plez, M. Bergemann, C. Evans, L. Patrick, "Red Supergiants as Cosmic Abundance Probes: Metallicities of M31," *The Astrophysical Journal*, 806, 21, 2015.
- P. S. Muirhead et al. (incl. **J. Z. Gazak**), "Kepler-445, Kepler-446 and the Occurrence of Compact Multiples Orbiting Mid-M Dwarf Stars," *The Astrophysical Journal*, 801, 18, 2015.
- C. Beichman et al. (incl. **J. Z. Gazak**), "Observations of Transiting Exoplanets with the James Webb Space Telescope (JWST)," *Publications of the Astronomical Society of the Pacific*, 126, 1134, 2014.
- M. Bergemann, R.-P. Kudritzki, M. Würl, B. Plez, B. Davies, **J. Z. Gazak**, "Red Supergiant Stars as Cosmic Abundance Probes: NLTE Effects in J-band Silicon Lines," *The Astrophysical Journal*, 764, 115, 2013.
- M. Bergemann, R.-P. Kudritzki, B. Plez, B. Davies, K. Lind, **J. Z. Gazak**, "Red Supergiant Stars as Cosmic Abundance Probes: NLTE Effects in J-band Iron and Titanium Lines," *The Astrophysical Journal*, 751, 156, 2012.
- B. J. Fulton et al. (incl. **J. Z. Gazak**), "Long-term Transit Timing Monitoring and Refined Light Curve Parameters of HAT-P-13b," *The Astronomical Journal*, 142, 84, 2011.
- J. N. Winn et al. (incl. **J. Z. Gazak**), "The Transit Ingress and the Tilted Orbit of the Extraordinarily Eccentric Exoplanet HD 80606b," *The Astrophysical Journal*, 703, 2091, 2009.

### **Selected Co-Authored Proceedings & Technical Reports**

- A. Cabello, J. Houchard, C. Harris, **J. Z. Gazak**, J. Kadan, J. Fletcher, "Learned Initial Orbit Determination from Simulated Electro-Optical Observations," AMOS Conference, 2024.
- M. Phelps, T. Swindle, **J. Z. Gazak**, J. Fletcher, A. Vandenberg, "Improving Spectral-Based Estimation of Space Object Orientation," AMOS Conference, 2022.
- M. Phelps, **J. Z. Gazak**, T. Swindle, J. Fletcher, I. McQuaid, "Inferring Space Object Orientation with Spectroscopy and Convolutional Networks," AMOS Conference, 2021.

Additional 30+ co-authored publications in ApJ, AJ, MNRAS, PASP, SPIE, and conference proceedings; see [Google Scholar](#) for the complete list.

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## Open-Source Software & Tools

**Sophon** — Procedurally generated benchmark for diagnosing spatial reasoning in vision-language models. Python package on PyPI (`sophon-bench`). [[github.com/zgazak](#) — private during review]

**AllClear** — Blind all-sky astrometric solver and directional sky-transparency mapper for autonomous observatory operations. [github.com/zgazak/allclear](#)

**SENPAI** — Sidereal Enriched Rate-Track Astrometry pipeline; sensor-agnostic, zero-calibration design for autonomous star/satellite detection. Python package on PyPI (`astro-senpai`). [github.com/ssc-ai/senpai](#)

**SDASim** — Speed-optimized differentiable satellite scene simulator for synthetic training data generation. PyTorch-based. [github.com/zgazak/sdasim](#)

**Burr** — Autonomous telescope calibration and observation scheduling system; integrates SENPAI and StarCSP for rapid-deploy SDA telescopes. [github.com/zgazak/burr](#)

**StarCSP** — Crowd-counting-inspired CNN for rapid star streak detection; trained with weak supervision from SENPAI ground truth on as few as one labeled streak per image. Released as GOTS.

**astroeasy** — Containerized astrometry.net plate solving with Python API. [github.com/ssc-ai/astroeasy](#)

**TAP** — Transit Analysis Package for exoplanet light curve analysis using Bayesian MCMC. Widely adopted in the exoplanet community; 100+ citations. [github.com/zgazak/TAP](#)

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## Peer Review & Professional Service

**Reviewer:** NeurIPS (2026), CVPR (2023–present), ECCV (2024–present), WACV (2021–2023), AMOS (2020–present)

**Session Chair & Invited Speaker:** 5+ invited talks at AMOS Conference; 4+ SPIE presentations; WACV oral presentation; numerous internal DoD conferences and stakeholder briefings.

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## Technical Skills

**Machine Learning & Interpretability:** Mechanistic interpretability (linear probes, attention attribution, Jacobian analysis, residual-stream interventions, activation patching), LoRA fine-tuning and placement grids, frozen capability audits, computer vision (CNNs, object detection, density estimation / crowd counting), Bayesian inference (MC dropout, SWA/SWAG ensembles), weak supervision, distribution shift robustness, synthetic data generation. PyTorch, scikit-learn.

**Vision-Language Models:** Architecture-level diagnostics across LLaVA-NeXT, LLaVA-OneVision, Qwen2.5-VL, Idefics3, InternVL2.5; frontier model evaluation (GPT-4o, Gemini, Claude); cross-architecture comparative analysis; encoder and decoder probe pipelines.

**Software & Infrastructure:** Scientific Python, distributed training (multi-GPU, 2× A6000 48GB), experiment management, LLM API integration and prompt engineering, full-stack devel-

opment (backend services, mobile, web), cloud infrastructure (AWS), serverless architecture, containerization (Docker), Git, CI/CD.

**Astronomical Instrumentation:** Optical/NIR spectroscopy, spectropolarimetry (LoVIS on AEOS 3.6m), telescope automation, astrometric calibration, plate solving, all-sky imaging, neuromorphic and multispectral sensor deployment. Observing experience on Keck (10m), Subaru (8.2m), IRTF (3m), VLT/KMOS (8.2m), AEOS (3.6m), and multiple 0.35-0.5m commercial telescopes.

**Leadership & Communication:** Technical strategy, cross-organization program management without direct authority, multi-stakeholder coordination (DoD, academia, industry), scientific writing, peer review for top-tier ML and astronomy venues.